



How to restore onboard key manager server configuration from the ONTAP boot menu

https://kb.netapp.com/on-prem/ontap/DM/Encryption/Encryption-KBs/How_to_restore_onboard_key_m...

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Applies to

- ONTAP 9
- Onboard Key Manager (OKM)

Description

Certain scenarios require onboard key manager server configuration to be restored prior to ONTAP fully booting.

Procedure

Preparation

The following detail will be necessary while restoring the OKM configuration:

1. Confirm the cluster is [using onboard key manager](#).
2. Obtain the cluster-wide passphrase that was entered while [enabling onboard key management](#).
3. Obtain the [backup information for the Onboard Key Manager](#).
4. [Verify onboard key management backup and cluster-wide passphrase](#).
5. Follow the table below to determine which combination of Boot Menu and ONTAP related steps are required.

Configuration Determination

Configuration Type	Process Steps Required	Command Verification of Configuration
Controller with node root volume encryption	Both the Boot Menu Steps and ONTAP Steps	Show the node root volumes and their encryption state: <code>cluster1::> volume show -is-cluster-volume false -encryption</code>
Controller with locked NetApp Storage Encryption (NSE) disks	Both the Boot Menu Steps and ONTAP Steps	Show encrypting drives without the default MSID (locked): <code>cluster1::> storage encryption disk show -data-key-id !0x0 and <code>cluster1::> storage encryption disk show -fips-key-id !0x0</code></code>
Controller with unlocked NetApp Storage Encryption (NSE) disks when Return a FIPS drive or SED to unprotected mode is	Only ONTAP Steps	Show encrypting drives with the default MSID (unlocked):

performed ahead of the maintenance

```
cluster1::> storage encryption  
disk show -data-key-id  
0x0 and cluster1::> storage  
encryption disk show -fips-key-  
id 0x0
```

Controller with [NetApp Aggregate Encryption \(NAE\)](#) and/or [NetApp Volume Encryption \(NVE\)](#)

Only ONTAP Steps

Show volumes with encryption enabled:

```
cluster1::> volume show  
-encryption-state !none
```

Boot Menu Steps

1. From the [ONTAP boot menu](#) select the appropriate option.

For **ONTAP 9.8 and later** select option **10**:

Please choose one of the following:

- (1) Normal Boot.
- (2) Boot without /etc/rc.
- (3) Change password.
- (4) Clean configuration and initialize all disks.
- (5) Maintenance mode boot.
- (6) Update flash from backup config.
- (7) Install new software first.
- (8) Reboot node.
- (9) Configure Advanced Drive Partitioning.
- (10) Set Onboard Key Manager recovery secrets.
- (11) Configure node for external key management.

Selection (1-11)? **10**

For **ONTAP 9.7 and earlier** enter the hidden option **recover_onboard_keymanager**:

Please choose one of the following:

- (1) Normal Boot.
- (2) Boot without /etc/rc.
- (3) Change password.

```

(4) Clean configuration and initialize all disks.
(5) Maintenance mode boot.
(6) Update flash from backup config.
(7) Install new software first.
(8) Reboot node.
(9) Configure Advanced Drive Partitioning.
Selection (1-9)? recover_onboard_keymanager

```

2. Confirm the continuation of the process.

```

This option must be used only in disaster recovery procedures. Are
you sure? (y or n): y

```

3. Enter the cluster-wide passphrase twice. While entering the passphrase the console will not show any input.

```

Enter the passphrase for onboard key management:
Enter the passphrase again to confirm:

```

4. Enter the backup information. Paste the entire content from the BEGIN BACKUP line through the END BACKUP line. Press the enter key twice at the end of the input. Truncated example below:

```

Enter the backup data:

-----BEGIN BACKUP-----
0123456789012345678901234567890123456789012345678901234567890123
[...]
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
-----END BACKUP-----

```

5. The recovery process will be completed.

```

Trying to recover keymanager secrets....
Setting recovery material for the onboard key manager
Recovery secrets set successfully
Trying to delete any existing km_onboard.wkeydb file.

```

Successfully recovered keymanager secrets.

```

*****
* Select option "(1) Normal Boot." to complete recovery process.
*
* Run the "security key-manager onboard sync" command to synchronize

```

the key database after the node reboots.

WARNING

If the output returns any result other than "Successfully recovered keymanager secrets." **do not** proceed. Perform troubleshooting to correct the error.

6. Select option **1** from the boot menu to continue booting into ONTAP.

* Select option "(1) Normal Boot." to complete the recovery process.

*

[...]

Selection (1-11)? **1**

7. Continue with the ONTAP Steps below.

ONTAP Steps

1. Confirm from the console the node reaches this state:

Waiting for giveback...(Press Ctrl-C to abort wait)

2. From the partner node issue a CFO only giveback:

```
cluster1::> storage failover giveback -fromnode local -only-cfo-  
aggregates true
```

3. Once booted only with the CFO aggregate execute the [security key-manager onboard sync](#) command:

```
cluster1::> security key-manager onboard sync
```

4. Enter the cluster-wide passphrase:

Enter the cluster-wide passphrase for the Onboard Key Manager:

All offline encrypted volumes will be brought online and the corresponding volume encryption keys (VEKs) will be restored automatically within 10 minutes. If any offline encrypted volumes are

not brought online automatically, they can be brought online manually using the "volume online -vserver <vserver> -volume <volume_name>" command.

5. If the sync is **successful** the cluster prompt is returned with no additional messages. If the sync **fails** an error message appears before returning to the cluster prompt. Do not continue until the error is corrected and the sync runs successfully.
6. Ensure that all keys are synced (no results should appear when filtering for **false** in the **restored** parameter:

```
cluster1::> security key-manager key query -restored false
```

There are no entries matching your query.

7. Finish the giveback of the node from the partner:

```
cluster1::> storage failover giveback -fromnode local
```

Additional Information

- [How to restore external key manager server configuration from the ONTAP boot menu](#)
- [Restore onboard key management encryption keys](#)
- [Onboard keymanager sync fails after motherboard replacement](#)
- [Supplied passphrase and backup data are not able to successfully import keys during maintenance](#)